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Exp no : 2

Experiment on Symmetric Ciphers

Understanding :

- A] This experiment focuses on substitution based symmetric ciphers, where encrypt/decrypt use the same key. Caesar cipher $\rightarrow$ shifts the letter by fixed position, Monoalphabetic $\rightarrow$ maps one letter to a letter, Playfair $\rightarrow$ it encrypts the text with help of digraph in 5×5 key matrix, Hill cipher $\rightarrow$ applies linear algebra in invertible matrix with modulus 26 as there are 26 char. Polyalphabetic $\rightarrow$ uses multiple shifting alphabets.
- B] For brute force, I needed to find out the secret key with given plain text and cipher text. For Caesar, I used ASCII values, so I generated all 95 possible shifts to check which pattern most likely would be used by user. Monoalphabetic would require frequency analysis. Playfair would require more sample of plain-cipher to get the digraph logic. Hill cipher checking invertible matrix would decode the key. Polyalphabetic was decoded by index of coincidence test.
- C] Application of these techniques was to secure messages in earlier days in security, military, communicating sensitive data, etc. The frequency graph shows letter occurrence probability in English, which attacker would exploit for encryption/decryption.
- Programming language : C++